IEEE ENICARthage SB

HU-ANOVATORS 2.0

MEDI DRONE

Autonomous Drone Delivery for Critical Medical Supplies



table of content

01 Introduction & Problem
02 The Solution Overview
03 Functionality
04 Implemented technology
05 Value & Proposition
06 BUSINESS MODEL CANVAS

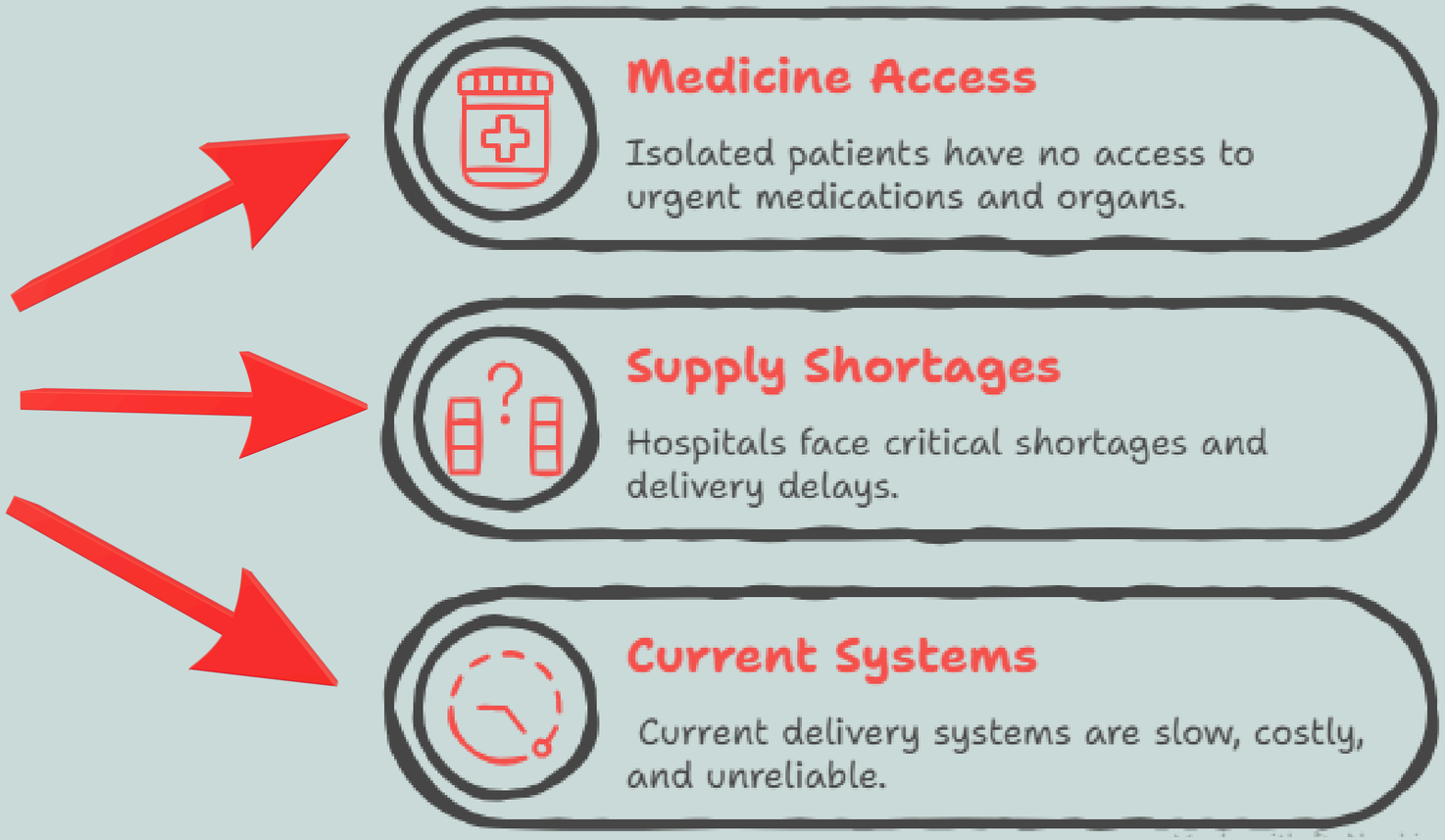
The Team



Introduction

5 million people die each year because medical care arrives too late — delayed by traffic, logistics, and distance.

Main problem



Our Solution

Drone delivery System coomponents



Drone-Based Delivery

A system for delivering medical supplies using drones.

Monitoring Dashboard

A dashboard for hospital staff to monitor deliveries in real-time.

Mobile App

An app for patients and staff to request and track deliveries.

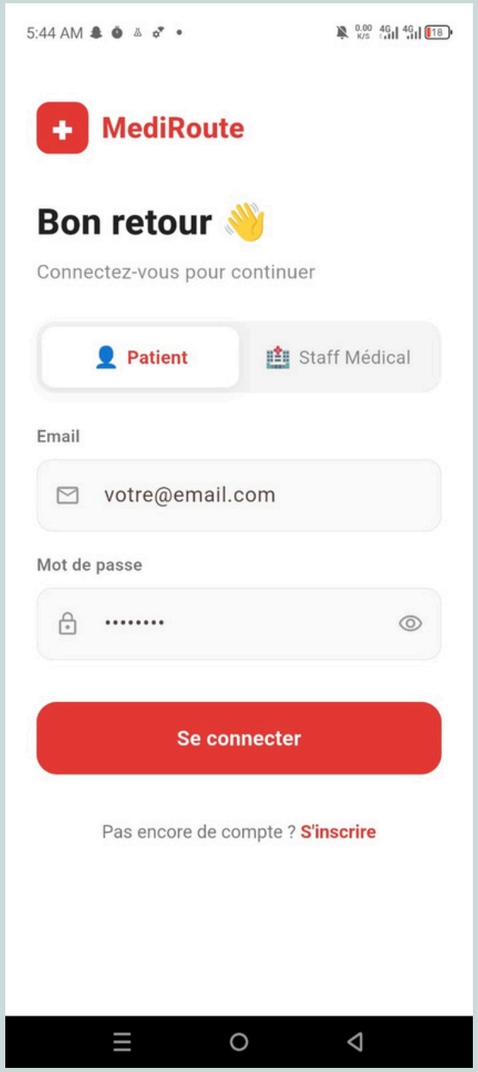
Autonomous Navigation

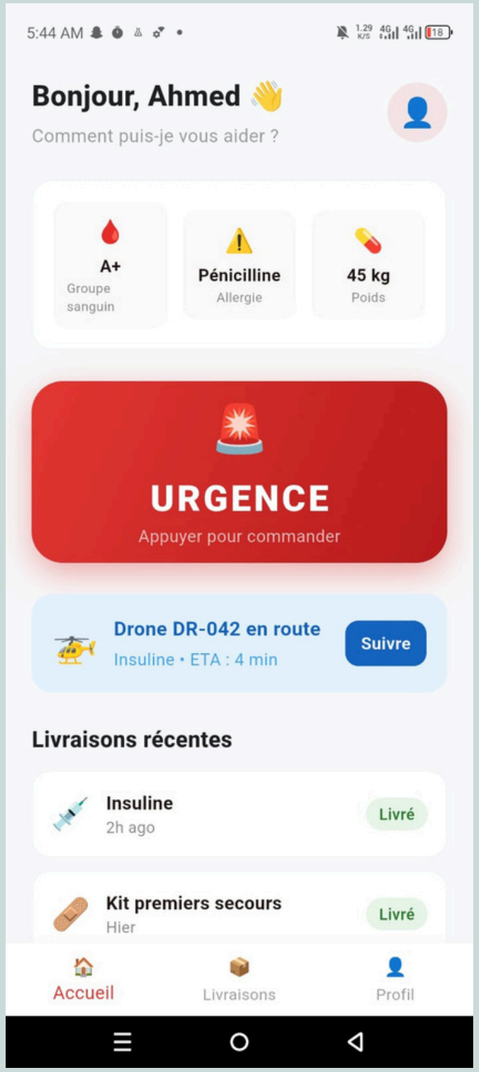
Navigation using ROS and PX4 for optimized delivery routes.

Delivery Focus

Focus on fast, reliable, and data-driven delivery, excluding blood.

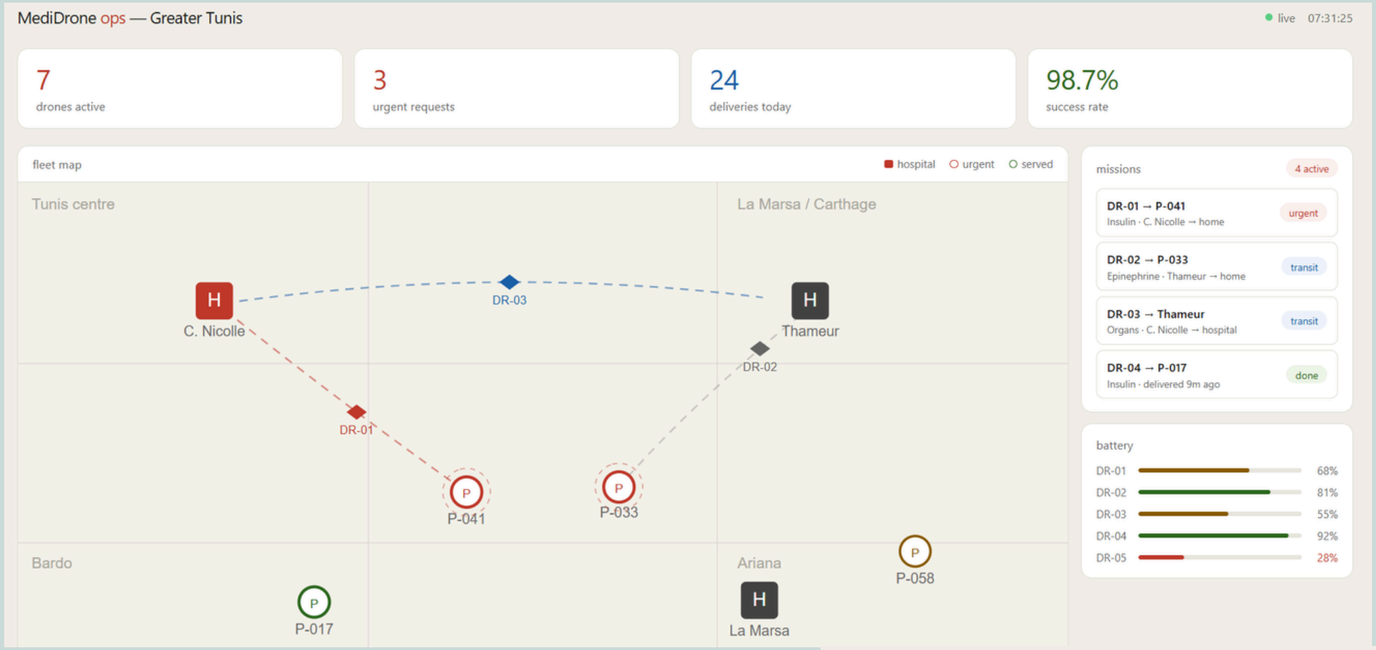
The App

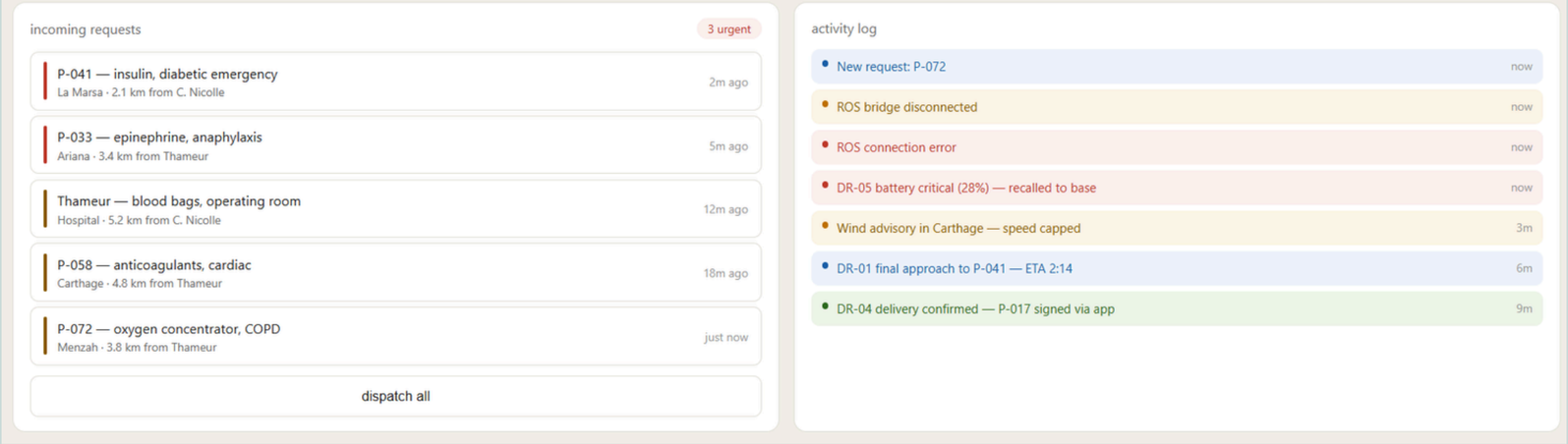

5:44 AM
MediRoute
Bon retour 👋
Connectez-vous pour continuer
Patient
Staff Médical
Email
votre@email.com
Mot de passe
••••••••
Se connecter
Pas encore de compte ? S'inscrire


5:44 AM
Bonjour, Ahmed 👋
Comment puis-je vous aider ?
A+
Groupe sanguin
Pénicilline
Allergie
45 kg
Poids
URGENCE
Appuyer pour commander
Drone DR-042 en route
Insuline • ETA : 4 min
Suivre
Livraisons récentes
Insuline
2h ago
Livré
Kit premiers secours
Hier
Livré
Accueil
Livraisons
Profil


5:44 AM
Que vous faut-il ?
Mode urgence activé
Catégorie de médicament
Insuline
Injection rapide
Cardiaques
Médicaments cœur
Premiers secours
Kit complet
Autres
Précisez ci-dessous
Note (optionnel)
Précisez votre besoin spécifique...
Localisation GPS confirmée
36.8189° N, 10.1658° E — Tunis Centre

The Dashboard





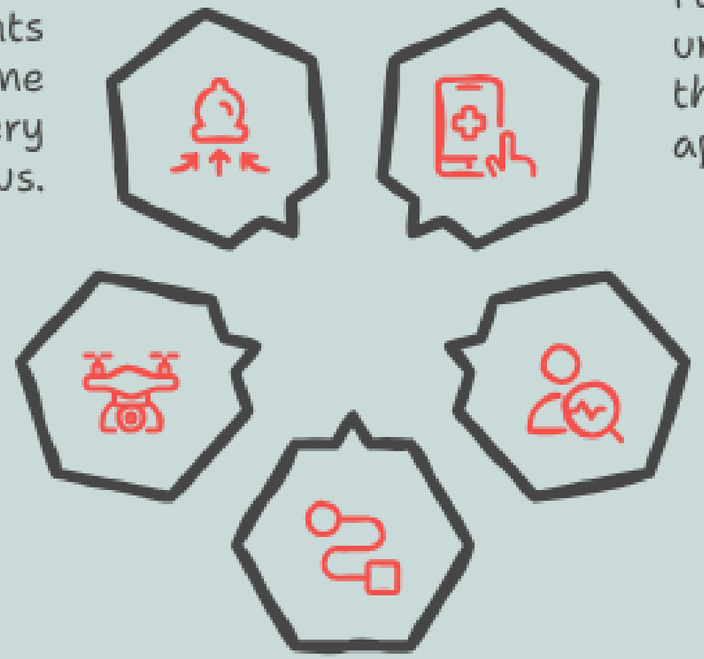
How It Works

Real-time updates

Staff and patients receive real-time updates on delivery status.

Patient request

Patient requests urgent delivery through the mobile app.



Drone flies

Drone flies autonomously to the location via PX4 autopilot.

System checks

System checks patient data, medical needs, and available drones.

Route optimized

Drone assigned and route optimized automatically using ROS.

Technology Stack

Gazebo

Realistic 3D simulation environment for comprehensive testin and scenario replication. Models physics, sensors, and complex environments.

PX4 Autopilot

Professional-grade open-source flight control stack ensuring precise navigation, mission execution, and fail-safe management.

The Tech Advantage: Dynamic Route Optimization

A* & Dijkstra

Foundational shortest-path algorithms providing robust, guaranteed baselines for initial route generation in static environments.

PSO Optimization

Advanced Particle Swarm Optimization for dynamic trajectory planning, minimizing time and energy while avoiding real-time obstacles.

Smart Routing

Context-aware navigation that synchronizes with real-time traffic flow, weather conditions, and emergency priority loads.

Value & Proposition

- Comprehensive Solution
- Technical Excellence
- Measurable Impact
- Human-Centric Innovation

BUSINESS MODEL CANVAS

Key Partners

- Hospitals and healthcare centers
- Drone suppliers and manufacturers
- Government and regulatory authorities
- Telecom providers (for connectivity & 5G support)

Key Activities

- Drone route optimization and management
- Delivery of urgent medical supplies
- Maintenance of drones and systems
- Software development and app management

Key Resources

- Drone fleet and equipment
- ROS-based navigation & optimization software
- Mobile app & web dashboard
- Cloud database with patient & hospital data

Value Proposition

- Fast and reliable medical delivery via drones
- Reduced emergency response time
- Access to healthcare in remote areas
- Real-time tracking of deliveries and drone activity
- Personalized delivery based on patient health data

Customer Relationships

- Real-time notifications and alerts
- Staff support via dashboard
- User-friendly mobile and web interface

Channels

- Mobile application (patients & staff)
- Web dashboard for staff and administration
- Direct integration with hospital systems

Customer Segments

- Hospitals & clinics
- Emergency centers (EMS)
- Patients with chronic diseases
- Rural healthcare facilities

Cost Structure

- Drone purchase and maintenance
- Software and mobile app development
- Cloud infrastructure and data security
- Operational costs for deliveries

Revenue Streams

- Annual patient subscriptions (similar to CNSS / insurance)
- Hospital subscriptions
- Pay-per-delivery fees
- Government contracts for emergency healthcare services

Thank you

For Your Attention

Q & A